

Lab 1

Veena Vadakkathmana Ramadas / 2024117008 / Group 13

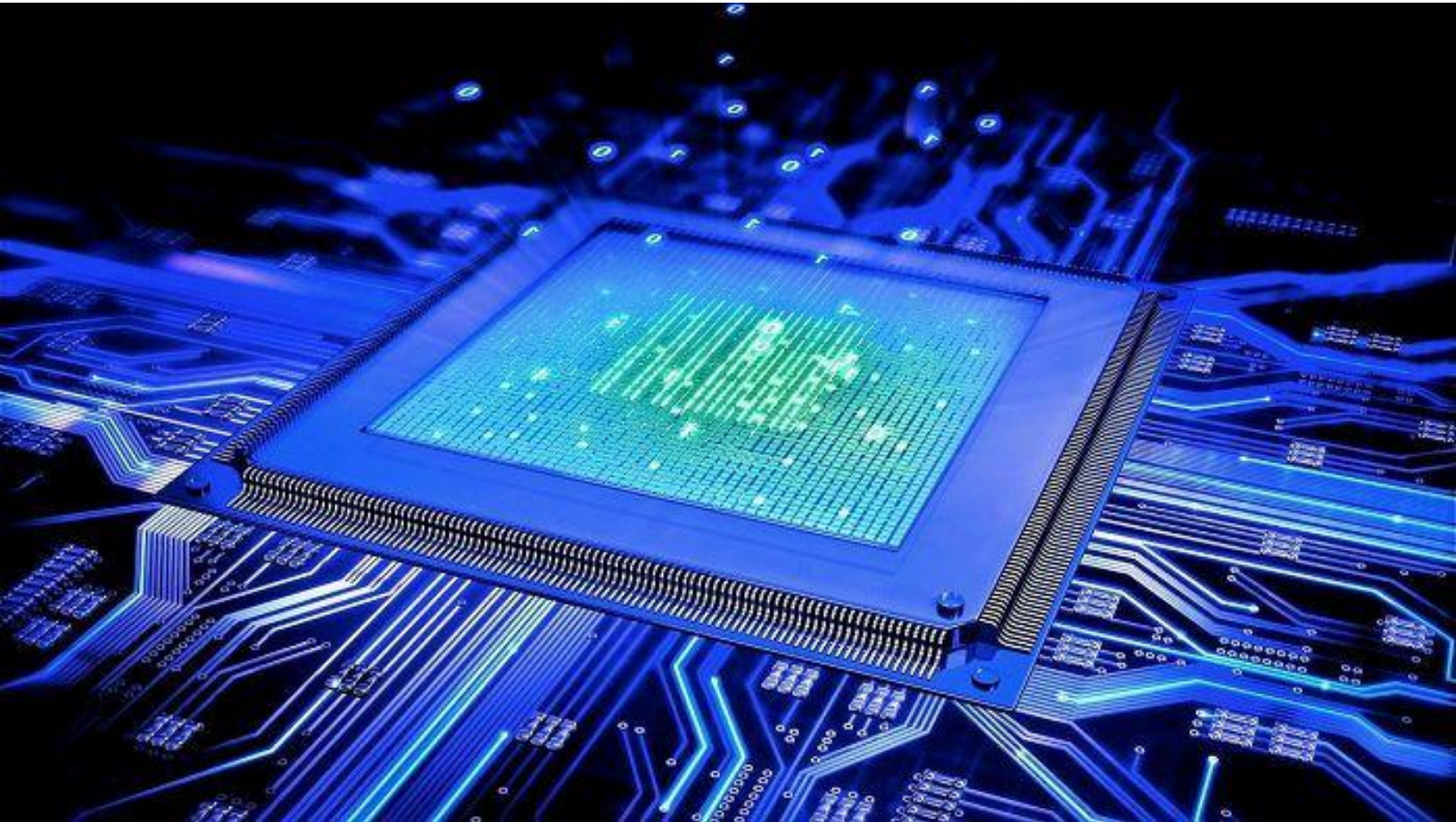
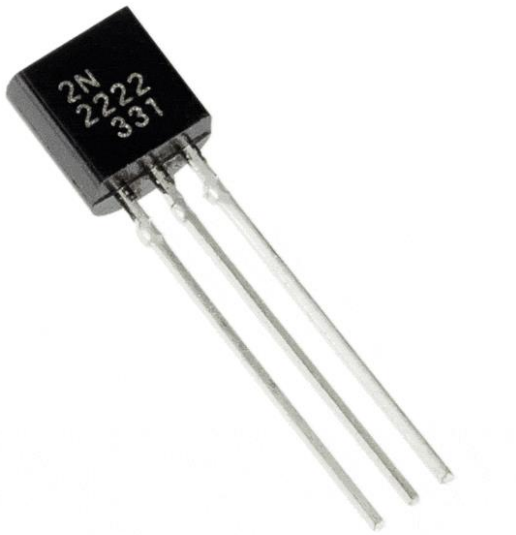


TABLE OF CONTENTS



Objective

Electronic components used

Reference circuit

Procedure

Conclusion

Link for Tinkercad

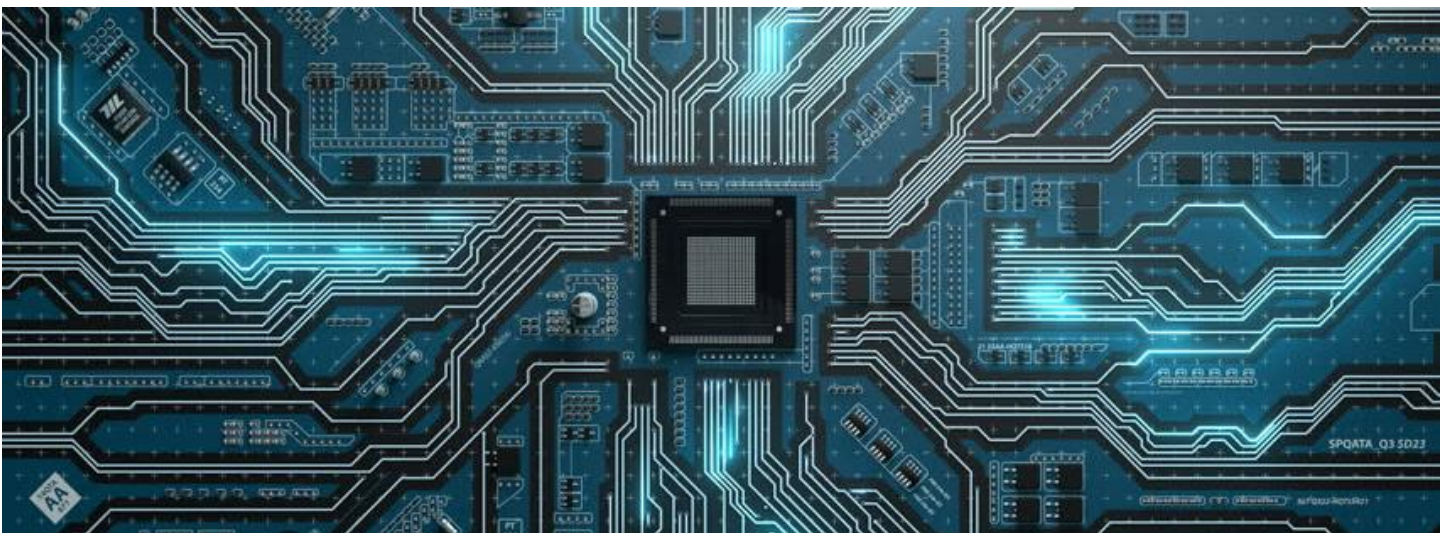
1A

Objective:

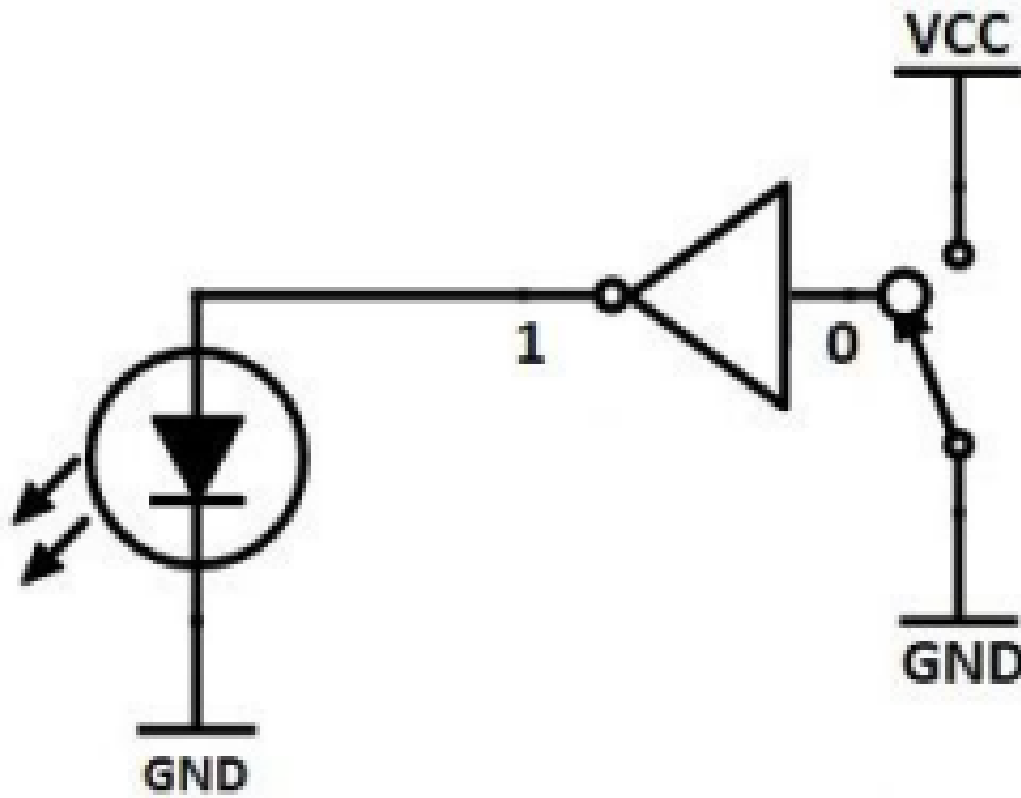
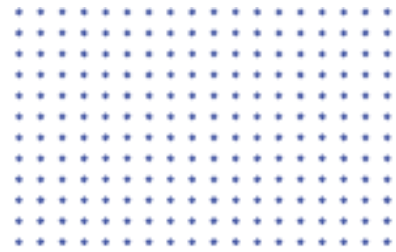
Get familiar with the use of Digital Test Kit and understand the use of NOT gate in digital circuits

Electrical components required

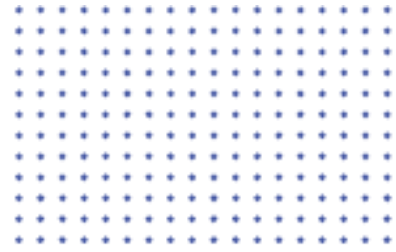
1. Breadboard
2. Input pins
3. LED
4. Resistor
5. IC (74HC04)
6. Wires
7. Power Supply



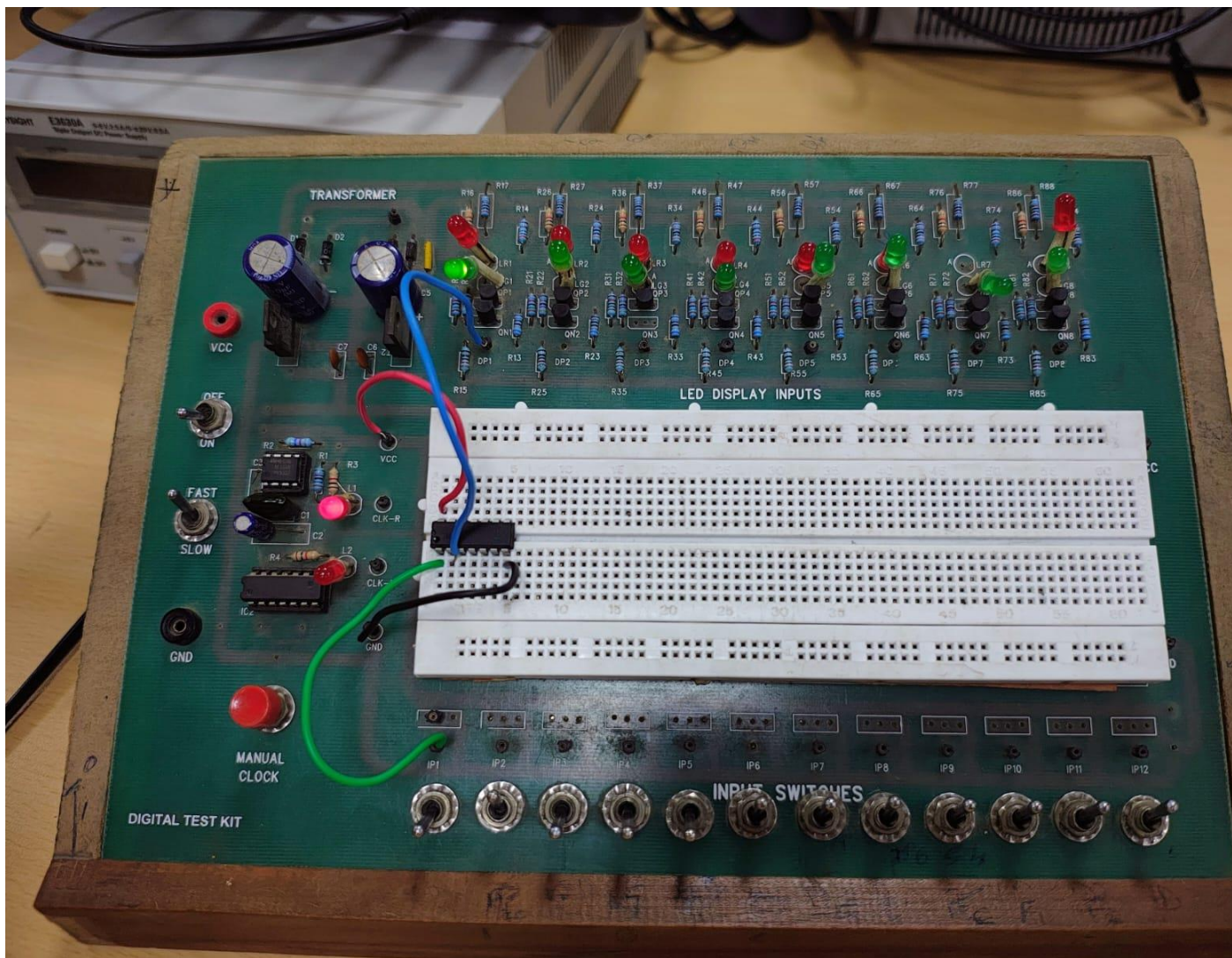
REFERENCE CIRCUIT



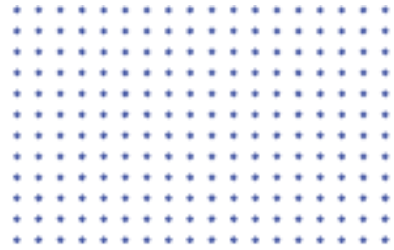
PROCEDURE



1. Get familiar with the schematic and the usage of the breadboard. Get a feel for the proper grip of the wire inserted into a tie-point.
2. Put the Clock control switch in the FAST position and switch on the VCC supply. Note that the CLK-R LED starts glowing. Using the Digital Multimeter in the DC VOLTAGE mode, measure and note the voltage between the VCC and GND pins.
3. Verify the working of all the Input Pins IP1-IP12 by measuring the voltages at these pins for both positions of the Input Switches. Verify the working of all the LED displays by applying an input from one of the Input Pins to DP1-DP8 and observing the LEDs as the Input is switched.
4. Implementation of NOT Gate - NOT gates are simply inverters. They simply invert the input logic for the output. Here we are going to use 7404 IC for demonstration. This IC has 6 NOT gates in it. These SIX gate are connected internally as shown in the figure.
5. Setup the circuit by the following steps
 - a. Connect the VCC and GND pins of the IC to the VCC and GND lines on the top and the bottom of the breadboard, using RED and BLACK wires respectively.
 - b. Connect the input pin of any one gate in the IC tone of the IP1-IP12 input switches, and the corresponding output pin of the IC to one of the DP1-DP8 display points provided in the Test Kit.



CONCLUSION

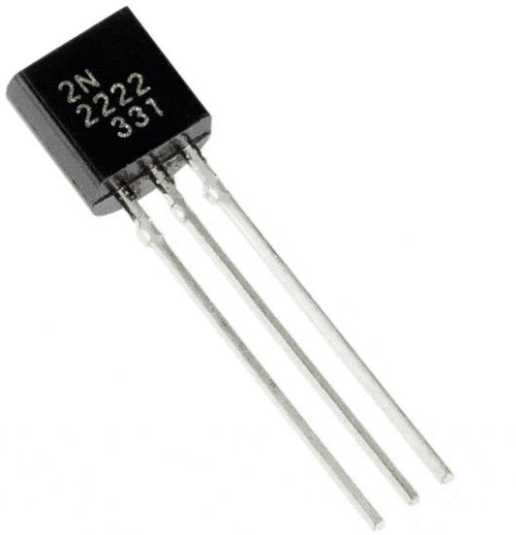


- NOT gate inverts the given input.
- Following is the truth table of NOT gate

Input	Output
0	1
1	0

Link for Tinkercad simulation: [Lab1_exp1](#)

TABLE OF CONTENTS



Objective

Electronic components used

Reference circuit

Procedure

Conclusion

Link for Tinkercad

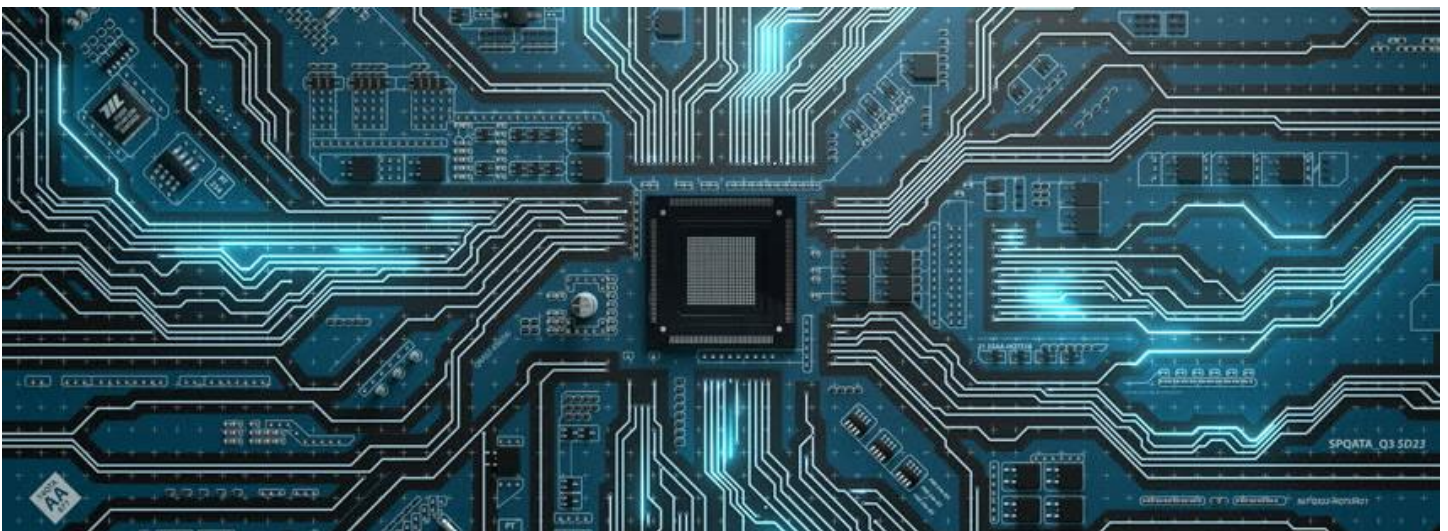
1B

Objective

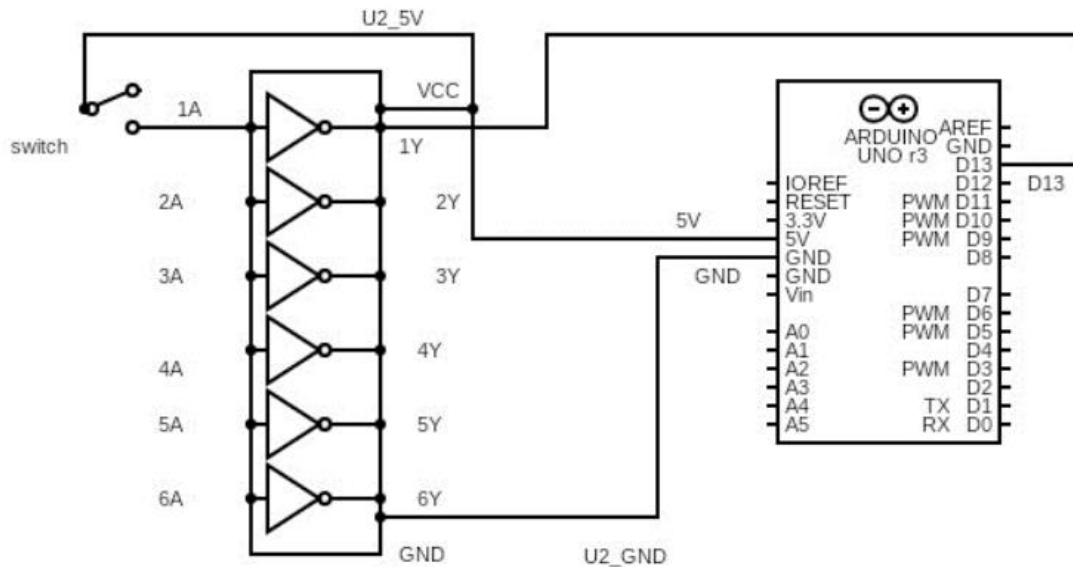
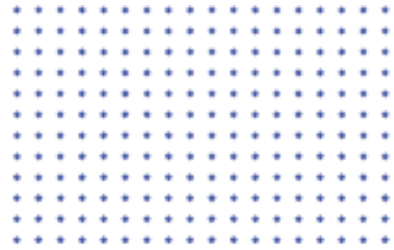
- Learn how to write and compile C code for ATmega328P in Arduino ISP.
- Dump the program into the microcontroller, run the program and observe its output.

Electrical components required

- Input switches
- Breadboard
- Resistor
- LED
- 74HC04 IC
- Wires
- Power Supply
- Arduino Uno 3

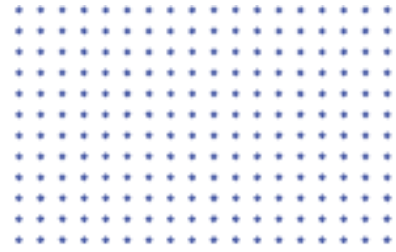


REFERENCE CIRCUIT

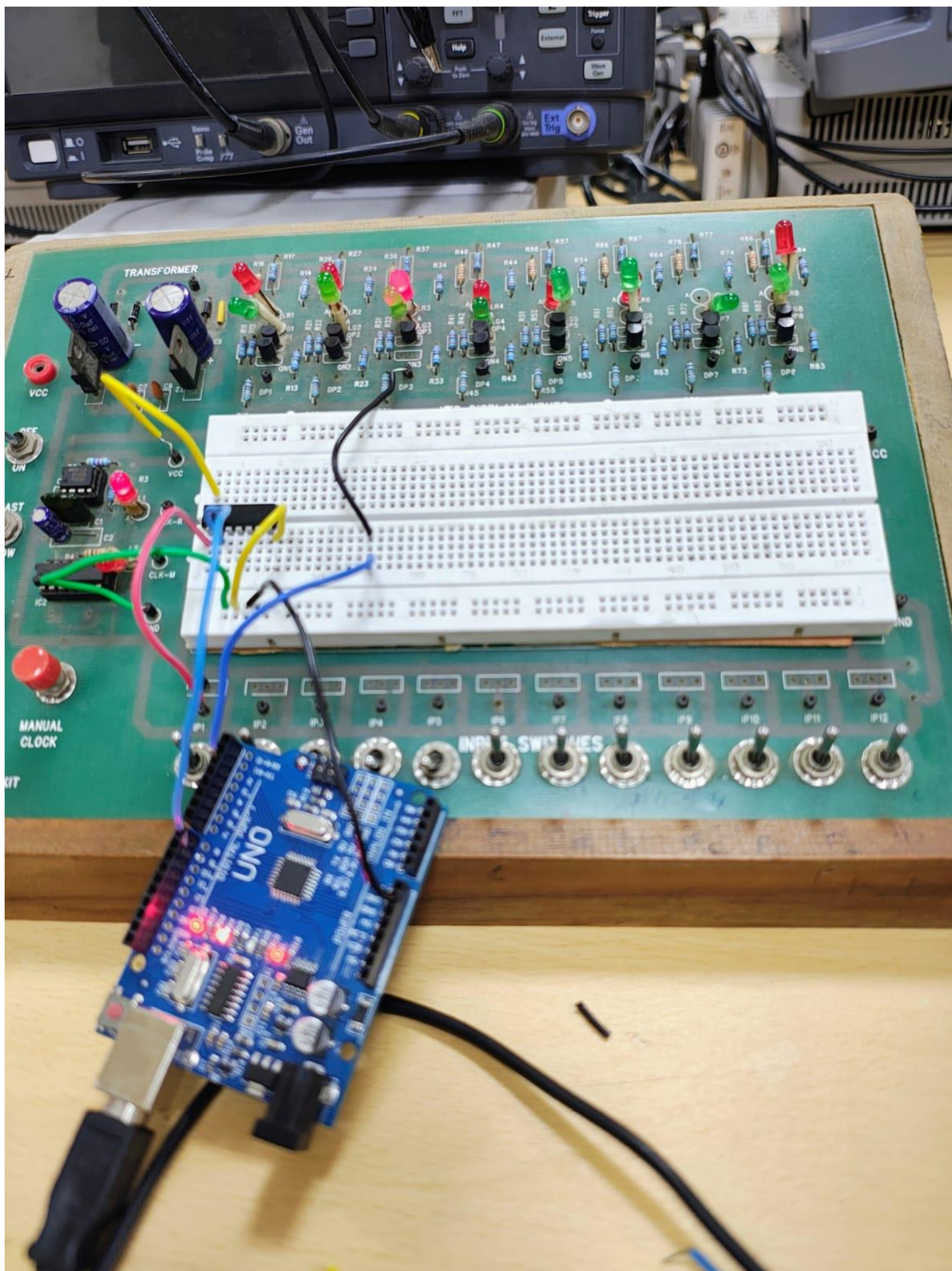


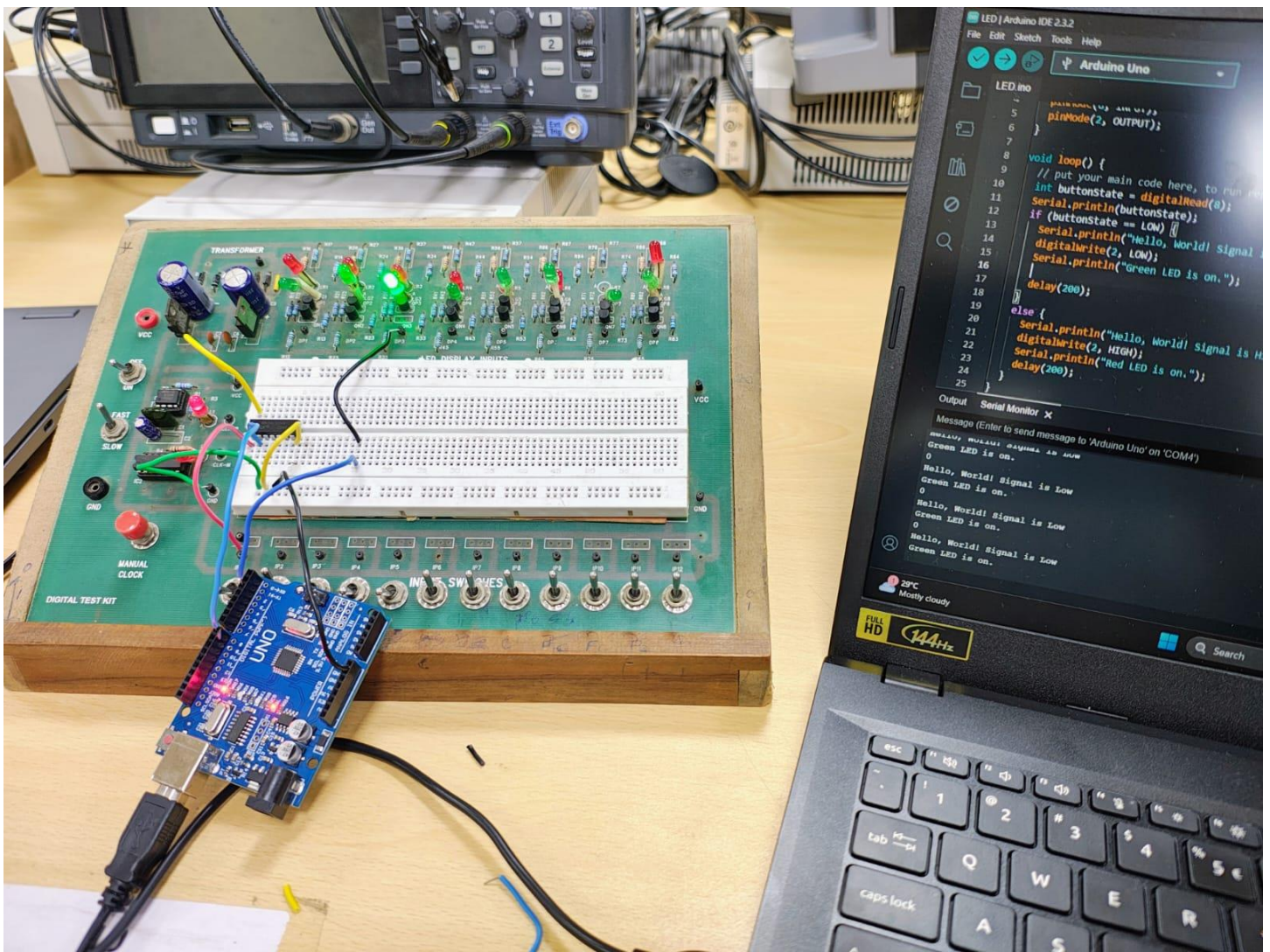
```
1 void setup() {
2   // put your setup code here, to run once:
3   Serial.begin(9600);
4   pinMode(8, INPUT);
5 }
6
7 void loop() {
8   // put your main code here, to run repeatedly:
9   int buttonState = digitalRead(8);
10  Serial.println(buttonState);
11  if (buttonState == LOW) {
12    Serial.println("Hello, World! Signal is Low");
13    Serial.println("Green LED is on.");
14    delay(200);
15  }
16  else {
17    Serial.println("Hello, World! Signal is High");
18    Serial.println("Red LED is on.");
19    delay(200);
20  }
21 }
```

PROCEDURE

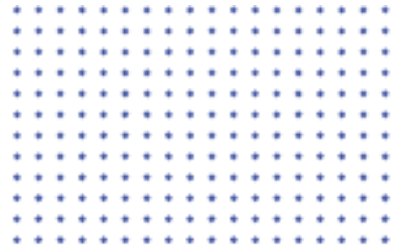


1. Download the provided code and open it with Arduino IDE.
2. Check the code for errors by compiling it.
3. In the 'Tools' section, choose the 'Board' type and select the correct 'PORT.'
4. Connect everything as needed and upload the code to the microcontroller.
5. Monitor the results on the Serial Monitor. Also, connect the circuit output from "Part A" to any of the 14 digital pins on the Arduino. Observe the output variations on the Serial Monitor as you test different input combinations.





CONCLUSION



NOT gate can be paired with Arduino to get the desired output.

Link for Tinkercad simulation: [Lab1_exp2](#)